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Blinkit Grocery Sales Analysis using Power BI and AI Tools

Institution – Graphic Era Deemed to be University (2026)

Project Domain

Power BI with AI

A Collaborative Initiative

Equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

Retail businesses like Blinkit face challenges in effectively analysing large volumes of sales data. The data is often complex, unstructured, and contains inconsistencies, making it difficult to identify trends and make informed decisions.

Without proper analysis, businesses may struggle with poor sales strategies, inefficient inventory planning, and a lack of visibility into customer behaviour.

This project addresses these challenges by using Power BI and AI tools to clean, analyse, and visualise data, enabling better decision-making and improved business performance.

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4. Objectives

List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

To understand the structure of the Blinkit dataset

To clean and preprocess the data using Power Query

To create key performance indicators (KPIs) using DAX

To analyse sales trends and product performance

To build interactive dashboards using Power BI

To generate insights using AI tools

5. Methodology

#	Phase	Description
1	**Data Collection & Exploration**	Collected Blinkit dataset (CSV/Excel) and understood data fields like sales, item type, and outlet details.
2	**Data Cleaning & Preprocessing**	Removed duplicates, handled missing values, and standardised data using Power Query
3	**Analysis & AI Integration**	Used DAX to create KPIs and analysed data. ChatGPT was used to generate insights and summaries.

4	**Visualisation / Output**	Created an interactive Power BI dashboard with charts, KPIs, and filters.
5	**Documentation & Submission**	Prepared project report, PPT, and demo video for final submission.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3
- Operating System: Windows 10
- Microsoft Power BI
- AI Chatbot ChatGpt

Internet connection — required for Copilot and AI Chatbot access

- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- Dataset: Blinkit Grocery Sales Dataset (Excel format)
- Contains fields like Item Type, Fat Content, Outlet Size, Location, Sales, and Ratings
- Data was cleaned using Power Query to handle inconsistencies and missing values
- Used for creating KPIs, analysis, and dashboard visualisation
- Helps in analysing sales trends and business performance.

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation	*Week 2*
3	Analysis, AI tool integration, and insight generation	*Week 2 – 3*
4	Dashboard/output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Conclusion

This project focused on analysing Blinkit grocery sales data using Power BI and ChatGPT. The data was cleaned and transformed, and important KPIs like total sales, average sales, number of items, and ratings were created. An interactive dashboard was developed to clearly visualise sales trends, product performance, and outlet analysis.

The key findings showed that some product categories generate higher sales, and outlet size and location play an important role in revenue. These insights help in understanding business performance more clearly.

All the objectives of the project were successfully achieved, including data cleaning, analysis, visualisation, and insight generation.

Overall, this project adds value by helping businesses make better decisions, improve sales strategies, and use data analytics and AI tools effectively in real-world retail scenarios.

8. References

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.